PES University — Department of Computer Science & Engineering

Lab 3: Requirements Engineering & RTM Deliverable

Problem Statement #55: Book Club Reading Challenge & Discussion Portal | **Domain:** Media, Events & Community
Student: Sushruth S | **SRN:** PES1UG24CS486 | **Section:** 4-H

1. Functional Requirements (FR)

Req ID	Requirement Name	Priority	Description & Acceptance Criteria
FR-001	Spoiler Masking & Reveal	**High**	Obscures `[spoiler]` comments with blur mask until clicked or chapter marked read. Search previews mask spoiler text. **Pass:** Text masked; reveals on click; search previews sanitized. *(Fixes BTBSC-1)*
FR-002	Reading Progress Tracking	**High**	Logs daily/monthly page counts, bounds check against total book length, updates progress %. Validates integer input. **Pass:** Increments accurately; rejects negative page numbers. *(Fixes BTBSC-2)*
FR-003	Discussion Moderation & RBAC	**High**	Discussion Leads can pin prompts, close threads, delete flagged posts. Regular members restricted. **Pass:** Leads perform moderation; members receive 403 Forbidden. *(Fixes BTBSC-3)*
FR-004	Democratic Monthly Poll	**High**	Allows exactly 1 vote per member per cycle. Atomic ballot commit and duplicate vote prevention. **Pass:** 1 vote recorded; subsequent submissions return 409 Conflict. *(Fixes BTBSC-4)*
FR-005	Milestone Badge Engine	**Medium**	Automatically awards profile badges upon reaching reading goals (100 pgs, 500 pgs, 1 book, streaks). **Pass:** Badge unlocked within 5s; no duplicate awards.
FR-006	Authentication & Profiles	**High**	User registration, Argon2id password hashing, stateless JWT session tokens, reading history. **Pass:** Signed JWT returned on valid login; invalid credentials return 401.
FR-007	Real-Time Poll Broadcast	**Medium**	Broadcasts live vote tally updates to all connected clients using WebSockets within 200 ms. **Pass:** Client charts update live without browser refresh.

2. Non-Functional Requirements (NFR)

Req ID	Quality Attribute	Target Standard / Metric	Architectural Mechanism & Acceptance
NFR-001	Performance & Concurrency	< 200ms P95 latency under 1,000 concurrent voters	Redis cache + connection pooling. **Pass:** Latency < 200ms; 0 transaction drops.
NFR-002	Security & Availability	99.9% uptime; AES-256 at rest; TLS 1.3 in transit	Automated health probes + column encryption. **Pass:** Zero plaintext PII; ≤ 44m downtime.
NFR-003	Usability & Accessibility	WCAG 2.1 Level AA; responsive 375px–1920px	ARIA labels, high contrast ≥ 4.5:1. **Pass:** Axe audit score 100%; responsive layout.
NFR-004	Data Consistency (ACID)	Strict atomic voting; compound unique index	PostgreSQL `UNIQUE(user_id, poll_id)`. **Pass:** Race condition duplicate attempts rejected.

3. Requirements Traceability Matrix (RTM)

Req ID	Use Case	Component / Subsystem	Jira Item	Pts	Test Case ID	Status
FR-001	UC-02, UC-07	Spoiler Filter & Masking Service	BTBSC-1, KB-101	5	TC-FR001-01, TC-FR001-02	**PASSED**
FR-002	UC-01	Reading Progress Engine	BTBSC-2, KB-102	5	TC-FR002-01, TC-FR002-02	**PASSED**
FR-003	UC-03	Discussion Moderation (RBAC)	BTBSC-3, KB-103	5	TC-FR003-01, TC-FR003-02	**PASSED**
FR-004	UC-04, 08, 09	Poll & Voting Engine	BTBSC-4, KB-104	5	TC-FR004-01, TC-FR004-02	**PASSED**
FR-005	UC-01, UC-06	Milestone Badge Service	KB-105, SC-205	3	TC-FR005-01, TC-FR005-02	**PASSED**
FR-006	UC-08	API Gateway & Auth Service	KB-106, SC-206	3	TC-FR006-01, TC-FR006-02	**PASSED**
FR-007	UC-09	WebSocket Live Tally Gateway	KB-107, SC-207	3	TC-FR007-01, TC-FR007-02	**PASSED**
NFR-001	UC-04, UC-09	In-Memory Redis Cache	NFR-101, SC-209	3	TC-NFR001-01 (1k Load)	**PASSED**
NFR-002	System-Wide	TLS 1.3 & AES-256 DB Layer	NFR-102, SC-210	3	TC-NFR002-01 (Cipher/Probe)	**PASSED**
NFR-004	UC-04	PostgreSQL Unique Transaction	BTBSC-4, NFR-104	3	TC-NFR004-01 (Race Test)	**PASSED**

4. Jira Defect Remediation Summary (Lab 2 → Lab 3)

Defect Key	Summary	Root Cause & Fix	Verification Method
BTBSC-1	Spoiler visible in search preview	Snippet generator leaked raw text. Added `sanitize_search_snippet()` to obscure `[spoiler]` tags.	TC-FR001-02 verified zero spoiler text in search output.
BTBSC-2	Negative page count accepted	Input lacked bounds validation. Added $0 < ext{pages} \le ext{remaining}$ guard in tracker.	TC-FR002-01 verified ValueError on invalid inputs.
BTBSC-3	Member accessed moderation controls	Missing RBAC claim check. Added `@require_role('DISCUSSION_LEAD')` enforcement.	TC-FR003-01 confirmed 403 Forbidden for non-leads.
BTBSC-4	Duplicate vote accepted on refresh	Tally updated before DB commit. Enforced composite constraint `UNIQUE(user_id, poll_id)`.	TC-FR004-02 confirmed 409 Conflict on duplicate attempt.